

Jialuo (Eric) Chen

☎ 365-996-2310 ✉ jialuo.chen@mail.utoronto.ca 📄 [Google Scholar](#) 🔗 [LinkedIn](#) 🐙 [GitHub](#) 🌐 [Website](#)

Education

University of Toronto

September 2022 – June 2027

Bachelor of Applied Science in Industrial Engineering with PEY-COOP, focus in AI & ML

Toronto, ON

- **Minors:** Computer Science, Artificial Intelligence in Engineering, Engineering Business | **Annual GPA: 3.62**
- **Thesis (MIE498Y1):** Role-Aware Bayesian Optimization under Uncertainty, advised by Prof. Justin Beland
- **Relevant Courses:** Machine Learning (MIE370, MIE424), Deep Learning (APS360), Artificial Intelligence (MIE369), Data Structures & Algorithms (MIE245), Databases (MIE353), Data Mining (MIE524), Decision Support Systems (MIE451), Reinforcement Learning (MIE369, MIE567), Software Design (CSC207, MIE350)

Research Experience

Data-Driven Decision Making Lab, University of Toronto

May 2026 – Present

Artificial Intelligence Research Assistant — Advisor: [Prof. Scott Sanner](#)

Toronto, ON

- Co-led development of **Embodied Immersive Conversational Recommendation System ([Demo](#))**, an AR headset application for studying how **persistent 3D world state** can ground conversational AI in real-world scenes.
- Developed an **object re-identification (ReID)** pipeline in **Python** combining RGB-D geometry and visual appearance to recognize the same physical objects across viewpoints while distinguishing lookalikes in cluttered scenes, achieving **98.3%** accuracy in tests involving 10+ shoes and chairs.
- Designed **asynchronous world-state updates** using versioned snapshots to reject stale results and keep recommendations tied to the correct objects across conversation turns, with stable test runs lasting **10+ minutes**.
- Co-designed the **software architecture** and led ReID/world-state refactoring, defining interface contracts to decouple core logic from perception and storage implementations and guide **human-AI collaboration** for long-term maintainability and future open-source contributions.

Li Shuai Research Lab, Shanghai Jiao Tong University

January 2026 – Present

Remote Deep Learning Research Assistant — Advisor: [Prof. Shuai Li](#)

Shanghai, China

- Conducted sustained research on **linear attention** for efficient context retrieval through kernel and memory-design experiments, alongside a separate **hidden-state steering** study on mathematical reasoning.
- Designed **attention kernels** for selective context retrieval and proposed a memory branch that maintains **context statistics**, evaluating it against kernel-only baselines through recall tests with distractors and language-model ablations on a **100M+** token subset of The Pile.
- Investigated **hidden-state steering** in **Qwen3.5-2B** by injecting representations from prior reasoning attempts to guide subsequent mathematical reasoning, comparing answer accuracy across injected prefix lengths up to **4,096 tokens** and different source layers.
- Exploring whether alternative **reconstruction objectives and optimization steps** could extend **Gated DeltaNet's expressivity**, building on its gradient-based formulation of memory updates.

JB Research Group, University of Toronto

January 2025 – May 2026

Machine Learning Research Assistant — Advisor: [Prof. Justin Beland](#)

Toronto, ON

- Proposed **Role-Aware Bayesian Optimization under Uncertainty (RA-BOUU)** through undergraduate thesis research, using input roles to guide modeling and experiment selection under execution and environmental uncertainty.
- Developed **multi-scale execution kernels** to capture local sensitivity and broader correlations across perturbed inputs, implementing analytic and semi-analytic marginalization to estimate expected deployment performance.
- Designed a **target-aligned acquisition rule** that selects environmental conditions for simulator evaluations by their expected reduction in uncertainty about average deployment performance under execution noise.
- Validated **RA-BOUU for simulated quadrotor control** in **Python** and achieved lower mean control cost and crash rate than four robust BO baselines across **three wind distributions** with **10 optimization runs** per setting.
- Extended BOUU with **multi-armed bandit portfolios** to adaptively select control and noise acquisition functions, achieving lower mean robust objective values than BOUU with a fixed noise acquisition function in preliminary Branin and Hartmann3 experiments.

Publications & Manuscripts

- **Jialuo Chen**, Haijing Wang, and Siyu Shao (2026). HMITL: Manager-Governed LLM Iteration with Guardrails and Rollback for Reproducible Healthcare Machine Learning Pipelines. *IEEE International Conference on Healthcare Informatics (ICHI)*, pp. 1218–1226. [Paper](#) | [Code](#)
- **Jialuo Chen** and Justin Beland (2026). Role-Aware Learning for Bayesian Optimization under Uncertainty. *Under review at NeurIPS 2026*.

Industry Experience

JANA Corporation

May 2025 – May 2026

Junior Software Developer

Aurora, ON

- Implemented a **Python reliability model** for large-load metering systems, combining precomputed failure-rate estimates through **Bayesian model averaging** and aggregating **12 threats** through fault trees to estimate leakage and supply-interruption probabilities.
- Delivered **full-stack improvements** to Lighthouse TIMP, a pipeline risk-management platform, building **React** and **TypeScript** confirmation flows and coordinating asynchronous UI states so compute-intensive risk simulations launch only after explicit user confirmation.
- Restored consistency between selected pipeline segments and exported datasets by correcting **C#/.NET** filtering logic with **LINQ** and validating the fix through **20+** regression tests.

Selected Projects

NeuroCommute [Code/Demo](#) | *Personalized Routing, A*, LLM Agent, Python*

December 2025 – April 2026

- Built **NeuroCommute**, a **natural-language route planner** that adapts Toronto transit, walking, and cycling recommendations to detailed preferences, including minimizing walking and prioritizing safety over travel time.
- Integrated TTC, GO, UP, and OpenStreetMap data covering **6,500+ locations** and **45,000+ connections**, using **A* search** with transfer and fare-window tracking to compare journeys by time, fare, and transfer burden.
- Implemented **critique-based recommendation** using LLMs to update routing weights and recompute ranked routes from feedback, retaining prior preferences so users can refine suggestions without restarting their search.
- Conducted a **92-respondent user study**, finding that **79%** had rejected map-suggested routes in the preceding two weeks, with walking, safety, and transfer concerns informing the planner's preference controls.

ADHD Scholarship Copilot [Code/Demo](#) | *React/JavaScript, Python/FastAPI, RAG*

November 2025

- Built a **Chrome extension** for students with ADHD, using the **Claude API** to turn dense scholarship instructions into concrete first steps so users can begin applications without planning the entire task upfront.
- Developed a **FastAPI** backend in **Python** that uses scholarship examples retrieved from **ChromaDB** to extract deadlines and reference requirements, giving applicants a concise summary of what to prepare and when to submit.
- Implemented **Chrome content scripts** with a docked task panel and cursor-following **Focus Shield**, helping users follow the next step while filling forms by keeping guidance visible and limiting visual distractions.

LineToLive Sketch-to-Face GAN [Code](#) | *Computer Vision, GAN, PyTorch*

June 2024 – August 2024

- Built a **three-stage conditional GAN** in **PyTorch** for sketch-to-face synthesis, separating grayscale reconstruction, colorization, and color refinement to supervise facial structure separately from color.
- Prepared **2,520 augmented sketch-photo pairs** and trained **U-Net generators** with a **PatchGAN discriminator**, combining L1 reconstruction and adversarial losses to preserve facial layout and encourage realistic local detail.
- Evaluated staged generation against a **single-generator Pix2Pix baseline** on held-out sketches, with qualitative comparisons showing sharper facial contours and fewer artifacts around face boundaries.

Skills

Programming: Python, C, Java, SQL, JavaScript/TypeScript, C#, Shell Scripting, R, MATLAB

AI/ML: Machine Learning, Deep Learning, NLP (Transformers, Linear Attention, Hidden-State Steering), Computer Vision (GANs, Object/3D Mesh Reconstruction), Generative AI (RAG, LLM Agents)

Probabilistic Methods & Planning: Bayesian Inference, Bayesian Optimization, Gaussian Processes, Multi-Armed Bandits, Reinforcement Learning (Q-Learning), MCTS, A* Search

ML & Data Libraries: PyTorch, scikit-learn, NumPy, pandas, Matplotlib, Seaborn, OpenCV

Application Development: Spring, React, HTML/CSS, FastAPI, .NET, LINQ, REST APIs, Chrome Extensions, PostgreSQL, ChromaDB, Spark

Research & Engineering: Software Architecture, Interface Design, Regression Testing, Model Evaluation, Ablation Studies, User Studies, Human-AI Collaboration

Tools & Platforms: Git, Docker, Linux, CUDA, JupyterLab, Jira

Professional Skills: Communication, Teamwork, Agile Development, Problem-Solving, Collaboration, Leadership

Honours & Awards

- Dean's Honours List**, University of Toronto — Winter 2026, Winter 2025, Fall 2024, Fall 2023.